

Department of Electrical Engineering

University of Moratuwa

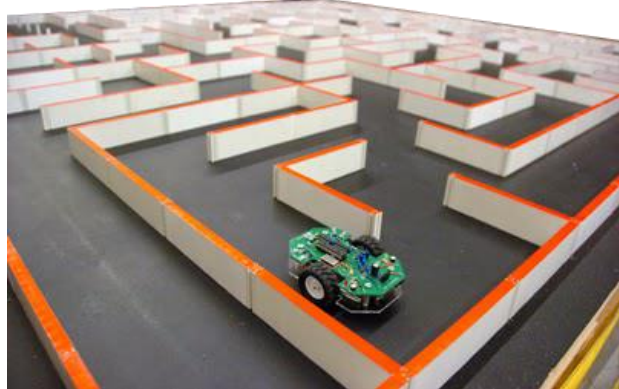
B.Sc. Eng. Semester 7

EE4360 - Embedded Systems Design and Programming

Maze Solver Challenge Project 2026

Maze solving is an event where a small autonomous robot solves a maze. It began in the late 1970s, and events are now held worldwide, especially in the UK, U.S., Japan, Singapore, India, and South Korea, while gaining popularity in subcontinent countries such as Sri Lanka.

In this year's challenge, the maze is divided into two connected sections. The first maze consists of a grid of cells, each 250 mm



square with walls 100 mm high. The robot begins from a predetermined starting position and must navigate through the maze to locate the bridge exit, placed at one edge. This bridge has an elevation of less than 35°, allowing the robot to climb autonomously.

A 3 cm wide black line runs across the bridge and continues into the second maze section. Students must implement line-following capability so that the robot can reliably cross the bridge and enter the new section. The second maze follows the same grid and wall dimensions as the first and contains the final target square.

The robot must therefore complete two shortest-path tasks:

1. Find the shortest route from the start point to the bridge exit in the first maze.
2. After crossing the bridge, find the shortest route from the bridge entrance in the second maze to the finish square.

The floor of the maze is black-colored, while the final goal square in the second section is marked with a white-colored floor. Having reached the finish, the robot may continue to optimize its path. Once the optimal route across both maze sections is discovered, the robot should execute this combined path in the shortest possible time.

Inspiring Video: <https://www.youtube.com/watch?v=ZMQbHMgK2rw>

MAZE SOLVER

The challenge is to build a small, self-contained autonomous robot capable of negotiating two connected maze sections in the shortest possible time. The robot will begin at the START tile in the first maze, navigate to the bridge exit, autonomously cross the bridge while following the black line, and then continue into the second maze section. From there, it must reach the FINISH tile in the shortest time. The robot that completes both maze sections, including crossing the bridge, and records the fastest time will be nominated as the winner.

The Robot

Construction:

- All teams should use the motors, wheels, motor driver, microcontroller, and sensors that are given by the university. Any team that uses alternatives for the above components will be immediately disqualified.
- The chassis of the robot must be 3D printed using the facilities available in the department. The design and construction must be primarily the original work of the team.
- The robot must be controlled autonomously with no human aid.
- The robot shall not leave any part of its body behind while navigating the maze.
- The robot shall not jump over, fly over, climb, scratch, cut, burn, mark, damage, or destroy the walls of the maze.
- The controller unit should be embedded in the robot and cannot be placed outside the robot.

Dimensions: The following size limitations apply to each robot

Width – maximum 18.5 centimeters

Length – maximum 18.5 centimeters

Height – 18 centimeters

Power Source:

- The robot must be powered by a power source such as a battery (15V Max) fixed on the robot. The Robot cannot be powered by a stationary power source connected to the robot by a cord.

The Field (Arena)

- Tiles, walls, and tolerances.

All grids use 25 cm × 25 cm tiles. Walls are 10 cm high and 1 cm thick (assume ±5% tolerance). Surfaces are on painted plywood; color uniformity and friction are not guaranteed. Floor seams between plywood sheets may snag low-hanging parts.
- Section sizes and enclosure.

The arena comprises two enclosed sections connected only by a bridge:

 - Section A (first maze): 4 × 4 tiles.
 - Section B (second maze): 9 × 9 tiles.

Outside walls fully enclose each section except at the designated bridge opening(s).
- Start tile (Section A).
 - The START is a full-white tile located at any tile position within Section A (not limited to corners).
 - At the start, the robot may be oriented in any direction, as instructed by officials before the run.
 - Timing begins when the robot's front edge first exits the boundary of the start tile into an adjacent tile.
- Bridge geometry and placement.
 - The bridge originates from a corner tile of Section A (the bridge-out tile) and enters a corner tile of Section B (the bridge-in tile).
 - The specific corners used, and the direction of the bridge, will be disclosed at the event; do not assume fixed compass directions.
 - Both the bridge-out and bridge-in tiles each occupy one full tile footprint to accommodate the connection through the outer wall.
- Approach the marker and line-following zone.
 - Approach marker: the tile immediately before the bridge-out or bridge-in tile contains a centered white square (9 cm × 9 cm) with a centered black square (3 cm × 3 cm), indicating that the next tile is a bridge ramp.
 - Surface treatment on bridge tiles: the bridge-out tile, the external bridge span, and the bridge-in tile have a white background with a black line of width 3 cm.
 - Post-entry line corridor: after leaving Section A, the white-background line zone continues for a short corridor and may include bends/turns before transitioning back to the standard maze surface (black floor, white walls). The exact corridor length and turns will be disclosed on the day.
- Finish tile (Section B).
 - The FINISH is a full-white tile located at any tile position within Section B (not restricted to corners).
 - Multiple paths to the finish are expected; the layout will be designed so that naïve wall-hugging alone is insufficient to guarantee discovery.
- General orientation and variability.
 - Start orientation, bridge corners/directions, the post-entry line corridor, and the finish location are variable and will be announced by officials.

- Actual layouts are subject to change before the event while remaining within the specifications above.
- Figures
 - Figure 1: Top view of both sections: Section A (4×4) and Section B (9×9) connected corner-to-corner via the bridge.
 - Figure 2: Tile level schematics of the bridge out and bridge in tiles (white background with 3 cm black line) and the approach marker tile (9 cm white square with 3 cm black square).

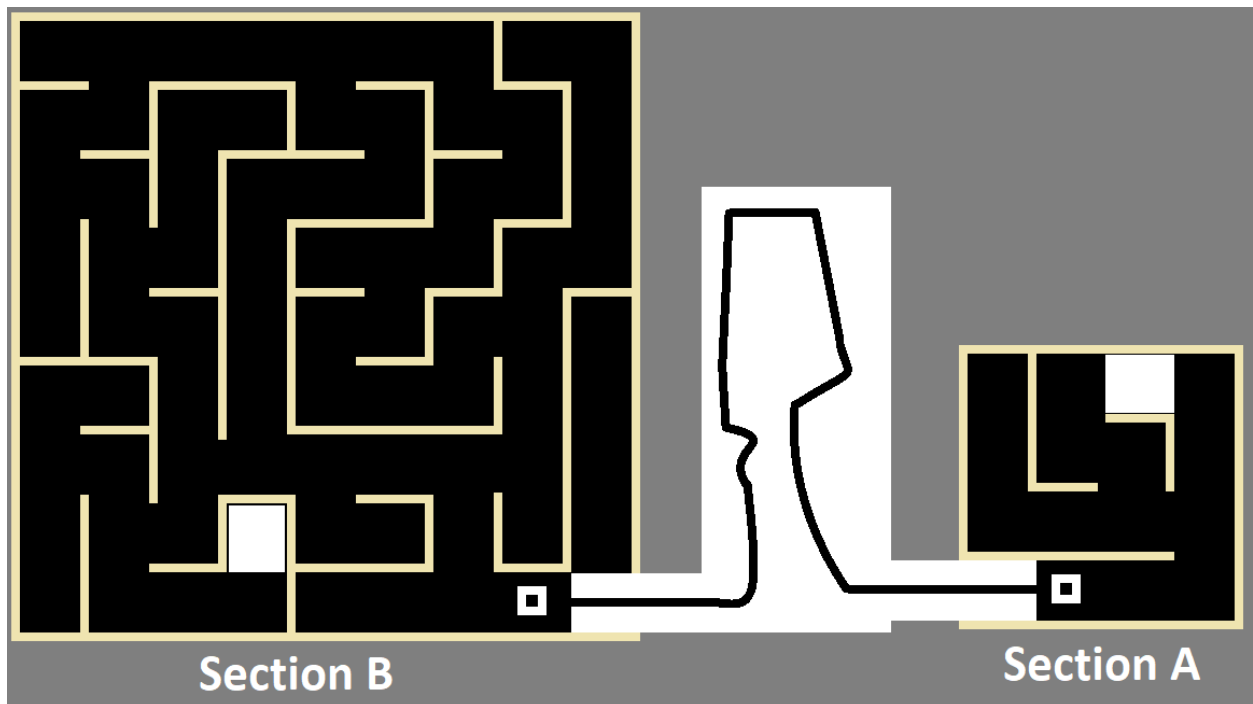


Figure 1

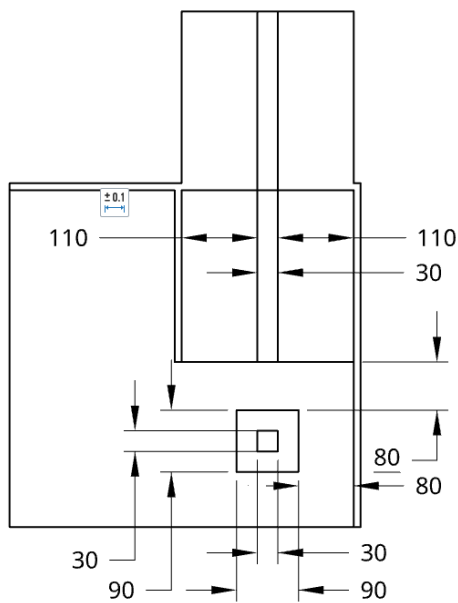


Figure 2

Gameplay

- **Race Clock.** The official race timer starts when the robot leaves the START tile and stops when it enters the FINISH tile. The elapsed time is recorded as the run time.
- **Trial Runs.** Each team is allowed a maximum of three trial runs. The best (fastest) run among these will be taken as the team's official time.
 - Each run must not exceed 8 minutes.
 - If the 8-minute limit is reached, a TIME-UP buzzer will sound, and the robot must be removed from the field.
 - If a robot is still actively navigating when the buzzer sounds, an additional 2-minute grace period will be given to allow the run to conclude without interruption.
- **Total Access Time.** Each team is allocated 30 minutes total access to the arena, including setup and adjustments between runs. The 30-minute clock continues running regardless of whether the robot is on the field or being adjusted.
- **Run Validity and Ranking.**
 - A run is valid only if the robot autonomously travels from the START tile to the FINISH tile.
 - The shortest run time across all trials is considered the team's official result.
 - Teams are ranked in order of fastest official time.
 - Robots failing to reach the FINISH are ranked based on the maximum number of consecutive tiles successfully traversed without human intervention.
 - If fewer than three robots finish within 30 minutes, the judges have discretion to assign rankings or declare no winners.
- **Abortions and Restarts.**
 - Each attempt begins when the robot exits the START tile.
 - If the robot re-enters the START tile before reaching the FINISH, that attempt is aborted and a new attempt begins.
 - If the operator touches the robot during a run, the attempt is immediately aborted.
 - If the robot has already completed a successful run, it may be removed without affecting the recorded time.
- **Multiple Runs.** After reaching the FINISH, the robot may continue exploring the maze as long as the total access time remains. However, time spent beyond the FINISH does not count toward the official run. A new official attempt begins only when the robot leaves the START tile again.
- **Timers.** Two timers are used:
 - A 30-minute total access timer per team.
 - A separate run timer for each attempt.
- **Permitted Adjustments During Trials.** Once the maze is revealed, no external information may be provided to the robot. However, teams may:
 - Change switch settings (e.g., left-turn vs. right-turn algorithm).
 - Replace batteries between runs.
 - Adjust sensors.

- Modify speed settings.
- Carry out necessary repairs.
- Downloading new programs or re-programming the robot during trials is strictly prohibited.
- Environmental Considerations. Teams must not assume consistent lighting conditions. Variations in sunlight, incandescent, or fluorescent lighting may affect sensor readings.
- Judging and Oversight. Judges may:
 - Request an explanation of the robot's operation.
 - Stop a run or declare disqualification if the robot breaks field integrity or violates rules.
 - Make final decisions regarding scoring, timing, and fairness.

Scoring

- The fastest-reaching robot to the finish line will be considered as winner. A maximum of 3 trials will be given to a team. The best time out of 3 trials will be considered. No human touch is allowed once the robot activation button is pressed.

Fair Play

- Robots that cause deliberate interference with other robots or damage to the field will be disqualified
- Students who cause deliberate interference with robots or damage to the field will be disqualified.
- It is expected that all teams aim to play a fair and clean game.

All decisions about scoring, gameplay play and timing are made by the juries. Teams should completely respect their vote and decisions. Members of the jury will be staff members.

Components Provided by the University

Component	Description	Quantity
Microcontroller	Microcontroller board (Arduino Mega) for controlling the robot.	1
Motors	DC Encoder motors for driving the wheels of the robot.	2
Wheels	Standard wheels compatible with the provided motors.	2
Wheel & Motor bracket	Metal bracket kit for motor and wheel fixing	2
Caster Wheel	Free-spinning caster wheel for balance and support.	2
Motor Driver	H-bridge motor driver module for controlling motor speed and direction.	2
Voltage Regulator	Voltage regulator to ensure a consistent power supply to the components.	1
ultrasonic sensors	Ultrasonic sensors for detecting walls and obstacles.	3
IR Sensor Array	8 sensor arrays to follow the line	1
Wires & Connectors	Standard wires and connectors for assembling the circuit. Male set and Female set.	2
Chassis (3D printed)	The chassis can be designed and 3D printed by the team using department facilities.	N/A
Gyroscope	Accelerometer Module Gyroscope MPU-6050 6DOF 3 Axis	1

- *If students require any additional equipment not listed above, they must obtain prior approval from the course instructor.*

Evaluation Criteria

1. Mid Evaluation (30 Marks) - **Deadline: TBD**

At this stage, the robot must demonstrate the ability to solve Section A and transition correctly towards the bridge, while also maintaining line-following skills.

Criteria	Description	Marks
Section A Navigation	The robot can navigate within Section A (4 × 4) to locate the correct bridge exit tile.	18
Line Following	The robot can reliably follow the line leading onto the bridge.	6
Autonomy and Stability	The robot demonstrates stable operation with no human intervention.	6
Total		30

2. Final Evaluation (70 Marks) - **TBD**

The final stage evaluates the robot's complete performance across the full arena: Section A, the bridge, and Section B up to the finish.

Criteria	Description	Marks
Section A + Bridge	Robot finds the shortest path to the bridge exit, correctly climbs the inclined bridge (<35°), and transitions into Section B.	15
Section B Navigation	Robot successfully navigates Section B (9 × 9) and reaches the finish tile.	15
Path Optimization	Robot demonstrates the ability to find and run the shortest path after exploration.	10
Autonomy and Stability	Robot operates reliably without human intervention; stable control and sensing	10
Design & Innovation	Creativity and effectiveness of mechanical design, including chassis and sensor placement.	10
Code Quality	Structure, readability, and efficiency of the embedded code.	10
Total		70

3. Bonus Points

- Completion Bonus section A (+5points): Awarded if the robot successfully finds the Bridge.
- Completion Bonus (+10 points): Awarded if the robot successfully solves the entire maze (Section A → Bridge → Section B → Finish).
- Ranking Points: Awarded based on the final standings:

- 1st Place: +35 points
- 2nd Place: +25 points
- 3rd Place: +15 points

Maximum attainable marks: 100 points.

Submission Requirements

Mid Evaluation (30 Marks) - **Deadline: TBD**

1. PlatformIO Project Files:
 - Submit the updated project folder reflecting progress toward Section A navigation and line-following improvements.
 - Code should be documented and structured according to good practices.
2. Short Video (2 - 4 minutes):
 - Demonstrate the robot successfully solving Section A and locating the bridge exit tile.
 - Show line-following performance at the bridge entry.
3. GitHub Evidence
 - Updated commit map screenshot reflecting progress since the initial evaluation.
 - Repository link confirming commits and pull requests.

Final Evaluation Submission (70 Marks) – **Deadline: TBD**

1. PlatformIO Project Files
 - Submit the complete and final PlatformIO project folder containing all code necessary for the robot's full functionality (Section A → Bridge → Section B → Finish).
 - The final version must reflect all improvements and be fully documented.
2. Short Video (5-10 minutes)
 - Demonstrate the robot solving the entire arena including Section A, the bridge crossing, and Section B to the finish tile.
 - Showcase advanced functionalities such as path optimization.
 - Explain overall design, control logic, and improvements from the mid evaluation.
3. GitHub Evidence
 - Final commit map screenshot showing the complete project development timeline.
 - Repository link confirming final commits and merges.

General Submission Instructions

- Only one member per group should submit the materials.
- Platform: All submissions must be uploaded to the course LMS as a compressed .zip file named after your team ID (Group_XX).
- File Contents: Each .zip must include:
 - PlatformIO project folder
 - Required demonstration video(s)
 - GitHub commit map screenshots and repository link
- Deadlines: Strictly follow the deadlines for Initial, Mid, and Final submissions. Late submissions will incur penalties.

Viva Examination

The viva is a compulsory individual assessment designed to ensure fairness and to confirm each student's real contribution to the project. Viva marks are given separately for each student, irrespective of the robot's overall performance in the competition.

Key Areas of Assessment

1. Code Understanding
 - Students will be asked detailed technical questions about the codebase, including algorithms, debugging methods, and optimization strategies.
 - Strict attention will be given to whether each student can explain not just *what* the code does, but also *why* it was written that way.
2. Commit Map Verification
 - During the viva, the instructor will open the GitHub repository and review the commit map and commit history.
 - Each student's claimed contribution must match the commit records (timestamps, descriptions, and files changed).
 - Consistency of contributions will be strictly checked — i.e., whether the student has contributed regularly and meaningfully throughout the project, rather than making minimal or last-minute changes.
3. Individual Contribution
 - Every student must clearly explain their own role and tasks in the project.
 - Students who cannot justify their contributions, or whose explanations do not align with the commit history, will receive reduced marks.
4. Whole-System Knowledge
 - Students are expected to demonstrate understanding of the entire robot system, not only the sections they personally implemented.
 - Questions may cover:
 - Section A navigation and bridge detection.
 - Line-following logic across the bridge.
 - Section B navigation and path optimization.
 - Interactions between hardware and software.
5. Hardware and Wiring
 - Questions will include the physical system: wiring, sensors, motor driver, microcontroller, and integration with the code.
 - Each student must be able to describe how hardware sections connect with and support the algorithms.

Marking and Fairness

- Each student will be awarded an individual viva mark out of 100%, based entirely on their own answers, understanding, and verified contributions.
- A student who fails to demonstrate understanding or whose contributions cannot be confirmed against GitHub records will receive reduced marks, even if their group robot performed perfectly in the competition.
- Conversely, a student who shows strong understanding and consistent contribution may receive a higher viva mark than their teammates.

Final Marks Calculation

Each student's Final Marks will be calculated from two components: the Team Robot Marks (T) and the Individual Viva Marks (V). The Bonus Marks (B) are added separately at the end and are not reduced by Viva performance. Maximum attainable marks: 100 points.

- T = Team marks (0–100), excluding bonus.
- V = Viva marks (0–100), individual.
- B = Bonus marks earned by the robot, added in full.

$$Final = \left(0.5 \cdot \frac{T}{100} + 0.5\right) \cdot V + B$$

Worked Examples

Team_X (Winning team)

- T = 85, B = 35
- Student AX (V=90): $(0.5 \cdot 0.85 + 0.5) \cdot 90 + 35 = 118.25 \rightarrow 100$ (capped)
- Student BX (V=75): $(0.5 \cdot 0.85 + 0.5) \cdot 75 + 35 = 104.38 \rightarrow 100$ (capped)
- Student CX (V=25): $(0.5 \cdot 0.85 + 0.5) \cdot 25 + 35 \rightarrow 58.13$

Team_Y (Completion only, Section B)

- T = 70, B = 10
- Student AY (V=100): $(0.5 \cdot 0.70 + 0.5) \cdot 100 + 10 \rightarrow 95$
- Student BY (V=65): $(0.5 \cdot 0.70 + 0.5) \cdot 65 + 10 \rightarrow 65.25$
- Student CY (V=30): $(0.5 \cdot 0.70 + 0.5) \cdot 30 + 10 \rightarrow 35.5$

Team_Z (Not completing)

- T = 60, B = 0
- Student AZ (V=85): $(0.5 \cdot 0.60 + 0.5) \cdot 85 + 0 \rightarrow 68$
- Student BZ (V=55): $(0.5 \cdot 0.60 + 0.5) \cdot 55 + 0 \rightarrow 44$
- Student CZ (V=45): $(0.5 \cdot 0.60 + 0.5) \cdot 45 + 0 \rightarrow 36$